PROJECT GUIDE

MissionMind

Evidence-Anchored, Graph-Augmented Retrieval for Cross-Mission Causal Analogy in Spaceflight Mishap Investigation Reports

Amaresh Nookala

Version 1.0 — 14 October 2026

Base paper: S. Kim and S. Kim, IEEE CASE 2026. Results in this guide come from a 10-mission pilot with AI-assisted labels; a fully human-labelled evaluation follows.

Contents

1. Executive Summary 3
2. Background and Problem 4
3. The Base Paper and Related Work 5
4. The Corpus 6
5. System Architecture 8
6. How the Project Was Built: Stages 9
7. Ingestion and Provenance 10
8. Reproducing the Base Paper 11
9. The Proposed Method (P) 12
10. The Causal-Factor Taxonomy 13
11. How the System Is Evaluated 15
12. Results: 10-Mission Pilot 16
13. Results: Per Mission, Generation and Verification 18
14. Review-Driven Improvements (P2) 19
15. Key Findings and Discussion 21
16. Limitations and Honest Scope 22
17. Project Status: Done and Remaining 23
18. How to Run the Project 24
19. Guide to the Remaining Human Labelling 25
20. Glossary 26
21. References 27

1. Executive Summary

MissionMind is a research system that helps spaceflight mishap investigators find *earlier accidents with similar causes* in long official investigation reports, and drafts a root-cause analysis grounded in the cited evidence. It reproduces a recent IEEE method for accident-report retrieval (Kim & Kim, IEEE CASE 2026) and adds an **evidence-verified causal-factor graph** on top of it.

What the project delivers

- A corpus of **27 documents** (23 investigation reports or summaries covering **22 missions**, 1967–2024, from NASA, ESA, JAXA and ISRO), split into **7,117 passages** that keep their exact page and character position.
- A faithful reproduction of the base paper's multi-field hybrid retrieval (BM25 + bge-m3 dense retrieval over three fields, fused with Reciprocal Rank Fusion), with every deviation documented.
- The proposed method **P**: an LLM proposes causal factors; each one is kept only if its quoted evidence is found word-for-word in the report (**2,067 of 2,830** kept); factors form a graph linking missions with shared causes; the graph is used as an extra retrieval channel; answers must cite evidence and may abstain.
- A complete evaluation and audit framework (metrics, significance tests, independent recomputation), a human annotation kit, and 162 automated tests.

Main results (10-mission pilot, AI-assisted labels)

- The proposed method P and the base-paper method B3 rank similar past missions about equally well: nDCG@5 **0.638** vs **0.648** (difference -0.010, 95% CI [-0.068, +0.049], p = 0.78) — **no significant difference**.
- The causal-factor graph used on its own (P-graphonly) gave the best ranking, nDCG@5 **0.713** — a promising but not yet significant signal.
- Verification rejected **27%** of the LLM's proposed factors, mostly because the "quoted" evidence does not exist in the report.
- After external reviews, a pre-specified second version **P2** (graph channel weighted equally with the text channels) reached nDCG@5 **0.678**; it also removed generic answers and reduced the text systems' bias towards long reports (chapter 14). Not significant on 10 queries.

Important disclosure. The pilot labels are **AI-assisted**: the project owner wrote all query descriptions and part of the causal-factor labels; the AI assistant (Claude) wrote the rest, using verbatim quotes from the investigation boards. A fully human-labelled benchmark will be completed after the project submission and before the IEEE paper submission. The results therefore show feasibility and trends, not proof that one method is better.

2. Background and Problem

When a spacecraft or launch vehicle fails, an investigation board writes a long report (often 50–300 pages) describing what happened, why it happened, and what should change. Many failures repeat causes seen before: the Columbia board explicitly compared its findings with Challenger seventeen years earlier. Yet finding such precedents today means reading many long reports by hand.

Why ordinary search is not enough

- **Different words, same cause.** "O-ring erosion accepted as normal" (Challenger) and "foam shedding accepted as an in-family event" (Columbia) share a cause — normalisation of deviance — but few words.
- **Long, mixed documents.** Reports mix background, chronology, analysis and recommendations; a keyword match in the wrong section is misleading.
- **Trust.** An investigator needs the exact page that supports every statement. Generative AI that invents sources is unacceptable in safety work.

The task MissionMind addresses

Cross-mission causal analogy: given a description of a new mishap, rank earlier missions whose *board-stated causes* are analogous, show the supporting passages with page numbers, and draft a root-cause analysis that cites only real evidence — or says "insufficient evidence".

Research questions

RQ	Question
RQ1	How reliable is evidence-anchored causal-factor extraction, and what recall does verification cost?
RQ2	Does a causal-factor graph channel improve cross-mission analogy retrieval over the base-paper method?
RQ3	What does the graph add beyond BM25, dense and hybrid retrieval (ablations)?
RQ4	Can answers be made verifiable at page level (citations, abstention)?
RQ5	How well do automatic metrics (LLM judge) agree with human judgement?

3. The Base Paper and Related Work

After comparing six candidate papers, the project selected **S. Kim and S. Kim, "Multi-Field Hybrid Retrieval-Augmented Generation for Maritime Accident Root Cause Analysis," IEEE CASE 2026, pp. 238–243, doi:10.1109/CASE69030.2026.11704480** as the base paper. It retrieves precedent accident cases from official investigation reports and drafts a grounded root-cause analysis — the same task family as MissionMind — and its method is fully specified and reproducible on local hardware.

The base method in one paragraph

Each accident report is split into three fields (Summary, Causes, Disposition). Each field is indexed twice: with BM25 (keyword ranking) and with the bge-m3 dense embedding model (meaning-based ranking). A query produces six ranked lists (3 fields × 2 methods), merged by **Reciprocal Rank Fusion**: score(c) = Σ 1 / (60 + rank). The top 40 chunks, labelled with case and field, are given to a large language model that writes a JSON root-cause analysis.

Closest related work

Work	What it does	Relation to MissionMind
Kim & Kim, IEEE CASE 2026	Multi-field hybrid RAG for maritime accident RCA	**Base paper** = baseline B3
Xiang et al., Aerospace 2026	Causal knowledge graph from aviation (NTSB) reports	Closest competing work; no retrieval task, no source links
Yang et al., Computers 2026 (AEVS)	Anchor-constrained, provenance-tracked LLM extraction	Inspires quote verification
Xie et al., Sci. Rep. 2025 (HybridRAG)	Graph + BM25 + vector for aircraft troubleshooting	Architecture precedent; not reproducible
Cormack et al., SIGIR 2009	Reciprocal Rank Fusion	Fusion used by B3 and P
Shappell & Wiegmann, 2000 (HFACS)	Human-factors accident taxonomy	Basis of the H and M taxonomy groups

Novelty statement. The retrieval technique itself is an adaptation. What is new is (1) the task and benchmark design for cross-mission causal analogy in spaceflight reports, (2) a causal-factor graph whose every edge is tied to a verified quote and page, and (3) a rigorous, audited comparison against a reproduced IEEE baseline.

4. The Corpus

All documents are official, publicly released investigation reports or summaries. Every file is recorded in `data/manifest.json` with its SHA-256 hash, size, page count, source address and role. Downloads were made only with the owner's approval, and each download was checked against the size recorded beforehand.

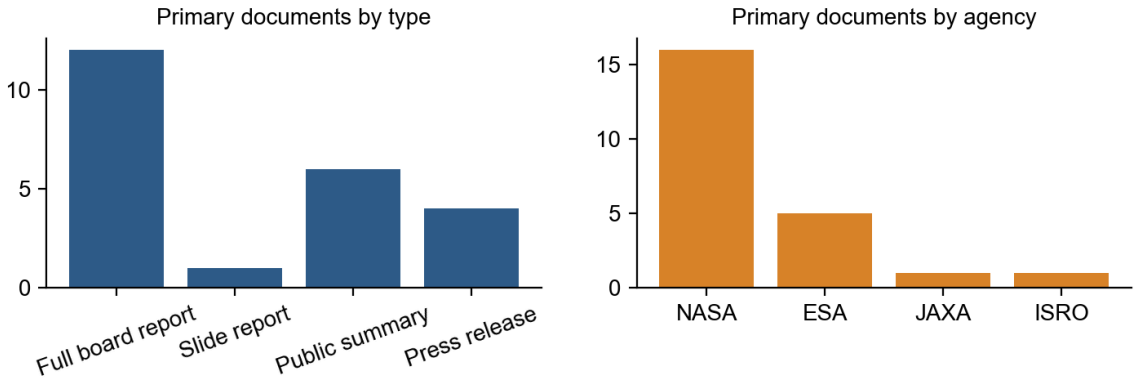


Figure 1. The 23 primary documents by type and agency.

Mission	Year	Agency	Type	Pages
Apollo 1 (AS-204)	1967	NASA	Full report	192
Apollo 13	1970	NASA	Full report	215
STS-51-L Challenger	1986	NASA	Full report	261
Ariane 5 Flight 501	1996	ESA / CNES	Full report	60
SOHO (mission interruption)	1998	NASA/ESA	Full report	31
Mars Climate Orbiter	1999	NASA	Full report	48
WIRE	1999	NASA	Full report	24
STS-107 Columbia	2003	NASA	Full report	248
NOAA-N Prime	2003	NASA	Full report	113
Genesis	2004	NASA	Full report	125
DART (2005)	2005	NASA	Summary	10
Orbiting Carbon Observatory (Taurus XL T8)	2009	NASA	Summary	6
Glory (Taurus XL T9)	2011	NASA	Summary	8
Antares Orb-3	2014	NASA	Summary	11
SpaceX CRS-7	2015	NASA	Summary	9
ExoMars 2016 Schiaparelli	2016	ESA	Full report	28
Hitomi (ASTRO-H)	2016	JAXA	Slides	104
OCO + Glory (Taurus XL T8/T9 update)	2019	NASA	Summary	8
Vega VV15	2019	ESA/Arianespace	Press release	1
Vega VV17	2020	ESA/Arianespace	Press release	1
GSLV-F10 / EOS-03	2021	ISRO	Press release	1
Vega-C VV22	2022	ESA/Arianespace	Press release	1

Boeing Starliner Crew Flight Test	2024	NASA	Full report	311

In addition, two ISRO annual reports are kept as *distractors* (not used in the main experiments) and two documents were *replaced* by better official versions: an ExoMars journal article (replaced by ESA's official inquiry report) and a university mirror of the Ariane 501 report (replaced by the official ESA copy).

5. System Architecture

MissionMind is a pipeline of small, separately tested Python modules. Everything runs locally: Neo4j 5 (graph database), Ollama with llama3 8B (language model), bge-m3 (embeddings) on an 8 GB GPU.

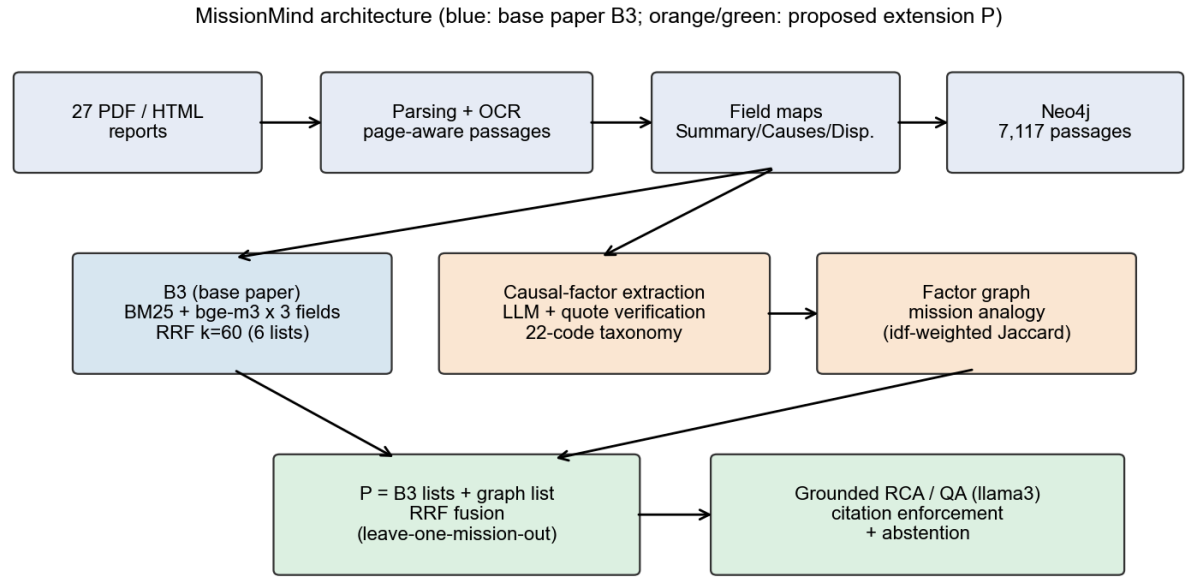


Figure 2. Data flow. Blue: reproduced base-paper method (B3). Orange/green: the proposed extension (P).

Folder	Contents
`src/pipeline/`	Page-aware PDF/HTML parsing, OCR, normalisation, field maps
`src/retrieval/`	BM25, dense retrieval, RRF, the base-paper systems (B1–B3), RCA generation
`src/extension/`	Causal-factor extraction and verification, factor graph, proposed method P, QA
`src/bench/`	Benchmark labels, metrics, statistics
`src/graph/`	Neo4j loading (passages, factors); Stage 1 system kept as baseline B4
`scripts/`	Runners, evaluators, audit, annotation tools
`runs/`	Every experiment: config, environment, predictions, metrics, logs
`docs/research/`	Stage reports 05–11

6. How the Project Was Built: Stages

The project followed a staged plan in which no stage was declared complete without evidence, and every run was saved in its own folder. Failures were kept and documented rather than hidden.

Stage	Goal	Outcome
1	Repair the original prototype; reproducible baseline	Broken retriever and evaluation repaired; fresh database; contaminated legacy graph preserved read-only
2	Ingestion and provenance	Every passage equals its page slice exactly; OCR quality measured (Challenger: text layer 14.4% vs OCR 3.3% character errors)
2b	Corpus expansion; fields	19 reports downloaded with approval; curated field maps for all reports
3	Reproduce the base paper	B0–B3 and the paper's comparison systems; byte-identical repeat runs; 13 deviations recorded
4	Proposed method	Verified extraction, factor graph, graph channel, cited generation, 6 ablations
5	Experiments	Evaluators and significance tests; synthetic proof of concept; 10-mission pilot
6	Audit	Independent recomputation, leakage, provenance and citation checks — PASS on all runs

Working rules followed throughout

- Never claim completion without verification; report failures openly.
- No fabricated results or citations; every number in this guide is read from a saved run file.
- Protect the evaluation set: no tuning on test labels; design changes are logged with date and reason.
- Preserve earlier results: runs are never overwritten; replaced items are archived, not deleted.

7. Ingestion and Provenance

The parser reads every page, detects headings by font size, weight and numbering, removes running headers and footers, repairs broken characters (ligatures, soft hyphens, line-end hyphenation) and splits text into passages of 150–300 words that never cross a page or section boundary.

The provenance guarantee

For every passage, `passage.text == page_text[char_start:char_end]`. A validator checks this for all 7,117 passages: **0 mismatches, 0 uncovered characters**. This is what allows every answer to point to an exact page.

Scanned reports and OCR

Several reports (Challenger, Apollo 1, Apollo 13, Genesis, the official Ariane copy) are scans. OCR quality was measured rather than assumed: for Challenger, the PDF's built-in text layer had a 14.4% character error rate against 3.3% for fresh Tesseract OCR, so scanned documents are OCR'd page by page. One report (Glory) has a text layer that silently cuts words ("Overvi SUMMA The Glo"); it is forced to OCR. The rotated French annex pages of the Ariane report cannot be read by OCR and are excluded.

Fields

The base paper's reports have native fields; spaceflight reports do not. Fields are therefore assigned by curated page-range maps written from each report's table of contents (heading keyword rules reached only 52–59% accuracy). Indexed passages per field: Summary 927, Causes 3,348, Disposition 189.

8. Reproducing the Base Paper

All the base paper's configurations were rebuilt and run on the same corpus and queries:

System	Description
B0	LLM only, no documents
B1	BM25 over all passages
B2	Dense (bge-m3) over all passages
H-whole	BM25 + dense, RRF, single index
SF-*	Single-field indexing (Summary only / Causes only), BM25, dense or hybrid
B3	**Base paper: 3 fields × 2 methods = 6 lists, RRF k = 60**
B3-bal	B3 with field weights ∝ √(field size) (added after a finding below)

Main deviations from the paper (13 recorded in docs/research/08)

Paper	MissionMind	Why
Native fields of short cards	Curated field maps over long reports	No native fields in mishap reports
BM25 parameters not stated	Okapi BM25, k1 = 1.2, b = 0.75	Standard defaults, not tuned
Train/test split	Leave-one-mission-out	Only 22 missions; each is query and precedent
Qwen3-Next-80B generator	llama3 8B (local)	8 GB GPU
Top 40 chunks in context	18–27 chunks (context limit)	8k-token window

Findings from the reproduction

- **Reproducible:** repeating a run gives byte-identical retrieval results.
- **Field imbalance:** with fields of very different size, RRF lets the small recommendation field dominate — 45% of B3's top-100 came from a field holding 4% of the text. Weighting fields by size overshot (100% Causes); √-size weighting gives a balanced mix (B3-bal).
- **Model nondeterminism:** llama3 did not reproduce its own answers across sessions (5 of 19 identical two hours apart), so every generated answer is cached and reported from the cache.

9. The Proposed Method (P)

Step 1 — Evidence-anchored causal-factor extraction

For each Summary or Causes passage, the LLM proposes causal factors, each with a taxonomy code, a role (root, contributing, observation) and a verbatim quote. A factor is **kept only if** the code and role are valid and the quote (at least 5 words) is found in the passage. The exact character position of the quote is stored, so every factor in the graph points to a page.

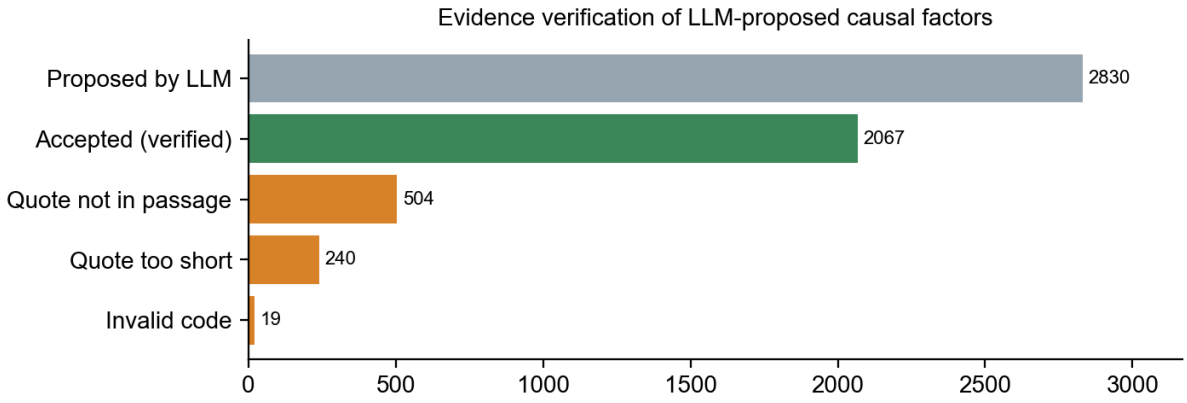


Figure 3. Verification kept 2,067 of 2,830 proposed factors (73%).

Step 2 — Factor graph

Verified factors are aggregated per mission into a profile of taxonomy categories and stored in Neo4j (22 missions, 2,038 factor mentions, 227 mission–category links). Two missions are analogous in proportion to the categories they share, rare categories counting more (idf-weighted Jaccard).

Step 3 — Graph channel

For a new mishap, the categories are taken from the query itself and from the verified factors in B3's top-20 passages (a description of symptoms rarely states causes). Missions are ranked by analogy, their evidence passages form a seventh ranked list, and this list is fused with B3's six lists by the same RRF.

Step 4 — Citation-constrained generation

The model drafts the analysis from the retrieved passages plus a short block of graph analogies. Afterwards, citations to passages that were not shown are removed, items without any valid citation are dropped, and if no grounded root cause remains the system **abstains** instead of guessing.

Ablations

Variant	What is removed or changed
P-noverif	Also uses factors that failed verification
P-nofields	Whole-index hybrid instead of the 3-field B3 lists
P-graphonly	Graph list alone
P-noidf	All categories weighted equally
P-noprf	Query categories from the query text only

10. The Causal-Factor Taxonomy

Factors are classified into 22 categories in five groups, based on NASA's HFACS human-factors levels and the technical failure classes used by the boards in the corpus. The taxonomy is a draft to be reviewed by the owner before full labelling.

Code	Category	Meaning
T1	Hardware design deficiency	design (geometry, materials, margins, redundancy) inadequate for the conditions; e.g. joint design, insulation, fairing separation design
T2	Software / requirements / specification error	flight or ground software fault, wrong requirement, unhandled exception, specification error
T3	Manufacturing / workmanship / material defect	defect introduced in production, assembly, handling or supplier processes
T4	Interface / integration / heritage-reuse mismatch	units mismatch, incompatible interfaces, component reused outside its qualified envelope
T5	Environmental condition beyond design	temperature, debris/foam strike, loads, radiation, atmosphere, thermal environment exceeding what the design tolerated
T6	Component failure (unexplained / wear / random)	part failed without a design or workmanship cause identified (e.g. thruster, valve, battery, gyro)
V1	Inadequate testing / qualification	tests missing, not representative, not end-to-end, qualification by similarity
V2	Inadequate analysis / modelling / simulation	models, simulations or analyses wrong, missing or not validated
V3	Inadequate review / independent assessment	design/peer/readiness reviews missed the problem; lack of independent verification
O1	Procedure deficient or not followed	operational/test procedure missing, wrong, or violated
O2	Operational configuration / parameter / data error	wrong command, parameter, uploaded data or configuration state
O3	Inadequate anomaly response / contingency planning	anomaly not recognised or handled; contingency plans missing or not executed
O4	Inadequate monitoring / telemetry / data use	available data not monitored, telemetry gaps, warnings missed
H1	Decision or skill-based error	an individual or team made a wrong decision or execution error
H2	Training / qualification / experience of personnel	personnel not trained, inexperienced, or unqualified for the task
H3	Communication failure	information not passed between people, teams, contractor and agency
M1	Schedule / cost / launch pressure	pressure to meet schedule or budget affected decisions
M2	Risk acceptance / normalisation of deviance	known anomalies accepted as normal; flawed risk assessment; flight rationale based on success
M3	Safety & mission assurance deficiency	safety/quality organisation weak, silent, not independent or absent
M4	Systems engineering / configuration management deficiency	requirements flow-down, configuration control, change management, systems integration process
M5	Oversight / organisational structure / roles	unclear responsibilities, inadequate contractor/government oversight or insight, matrix management issues

M6	Staffing / resources	teams understaffed or spread thin, insufficient resources

11. How the System Is Evaluated

Benchmark sets (annotation kit in evalutation/benchmark/)

Set	Content	Used for
D2a	Board-stated causal factors per mission (code, role, page, quote)	Defines which missions are analogous
D2q	A 100–200-word description of each mishap as first observed, without causes	The test queries
D2b	Analogy grade 0–3 for every pair of missions	Ranking relevance
D1	Causal factors in 200 sampled passages	Extraction accuracy (RQ1)
D3	About 105 questions with key points and evidence pages	Answer quality, abstention

Metrics

- **nDCG@5** (primary): quality of the top-5 mission ranking, rewarding highly analogous missions near the top.
- **Recall@5, MRR**: share of analogous missions in the top 5; position of the first analogous mission.
- **NormRecall@100, nDCG@100**: the base paper's passage-level metrics.
- **Extraction precision/recall/F1**; **key-point recall**, **citation precision** and **abstention accuracy** for answers.

Statistics

One pre-registered primary test: P vs B3 on nDCG@5, with a paired permutation test and a 95% bootstrap confidence interval. All other comparisons are secondary and corrected for multiple testing (Holm). The protocol was written before any label existed (docs/research/09).

Independent audit

A separate script, sharing no code with the evaluator, checks that every system answered the same queries from a clean code version, that no query leaks into its own answers, that every retrieved passage matches its page, that every citation is valid, and recomputes the metrics.

12. Results: 10-Mission Pilot

AI-assisted labels. MIXED: owner (D2q for all 10 missions; D2a factors and quotes for Columbia, Genesis, SOHO, WIRE, Starliner) + assistant Claude (taxonomy codes/roles/page fixes for owner rows; D2a for Challenger, MCO, Ariane 501, Apollo 13, Hitomi). Not a human-only benchmark. Every label row records its author. The AI assistant also built the method under test, so these labels may favour it. Fully human labels will follow.

Missions: Challenger, Columbia, Mars Climate Orbiter, Ariane 501, Apollo 13, Genesis, SOHO, WIRE, Hitomi, Starliner. Each query is the owner's description of one mishap; the system ranks the other 9 missions.

System	nDCG@5	nDCG@10	Recall@5	MRR	NormRecall@100
B1	0.644	0.831	0.535	0.833	0.022
B2	0.644	0.806	0.571	0.733	0.047
H-whole	0.658	0.826	0.557	0.850	0.030
SF-summary-hybrid	0.675	0.835	0.564	0.833	0.039
SF-causes-hybrid	0.642	0.828	0.515	0.900	0.039
B3	0.648	0.800	0.580	0.733	0.019
B3-bal	0.642	0.828	0.515	0.900	0.035
P	0.638	0.816	0.540	0.833	0.023
P-noverif	0.672	0.831	0.555	0.900	0.019
P-nofields	0.630	0.804	0.540	0.783	0.040
P-graphonly	0.713	0.860	0.598	0.883	0.056
P-noidf	0.638	0.816	0.540	0.833	0.023
P-noprf	0.638	0.811	0.555	0.783	0.022

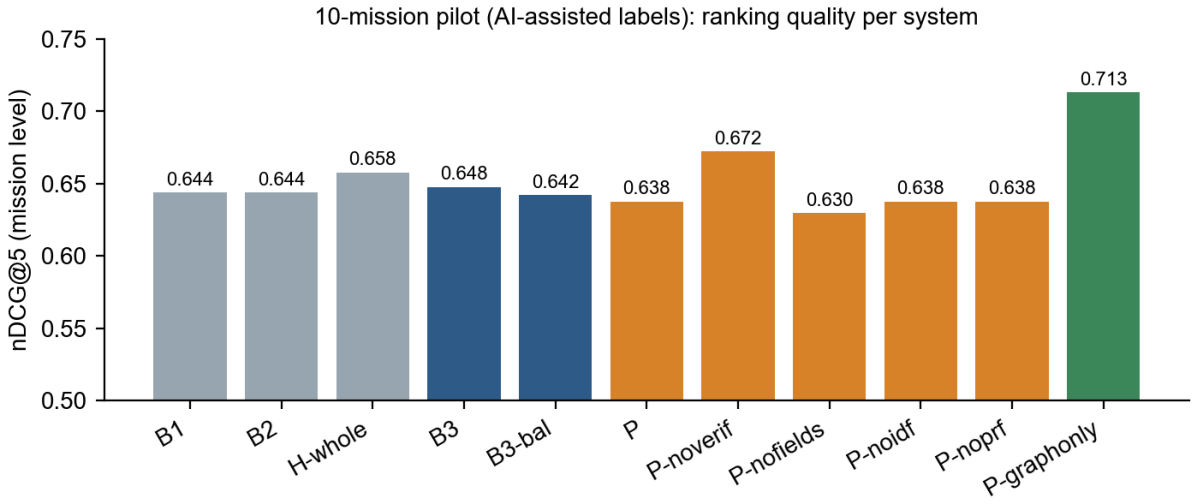


Figure 4. Pilot ranking quality (nDCG@5). Grey: simple baselines; blue: base paper; orange: P and ablations; green: graph only.

Primary comparison

P − B3 = **-0.010** nDCG@5, 95% CI [-0.068, +0.049], exact permutation p = **0.78** (n = 10). **No**

difference was detected. Of 62 secondary comparisons, 0 are significant after Holm correction.

13. Results: Per Mission, Generation and Verification

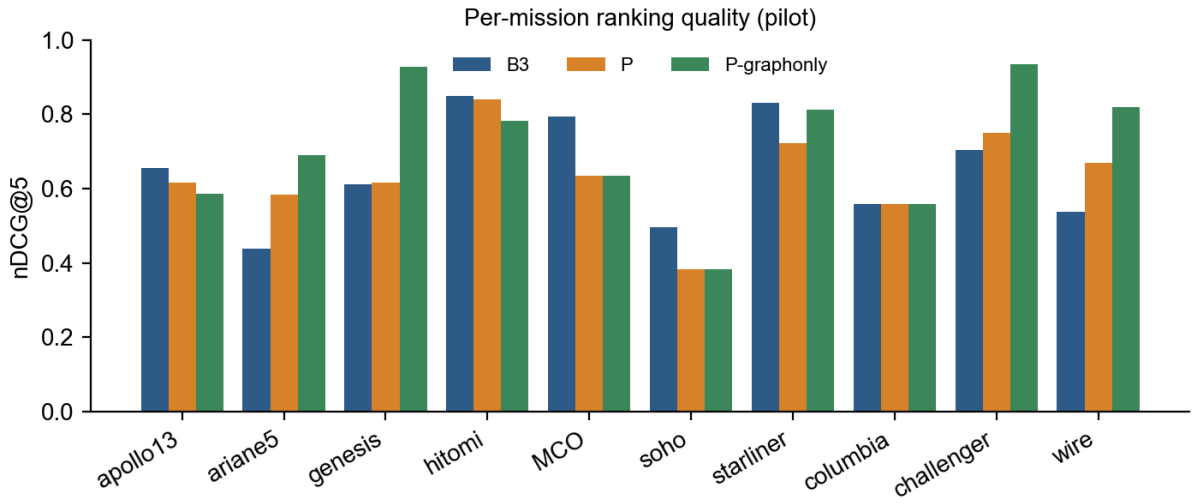


Figure 5. Per-mission nDCG@5 for B3, P and the graph alone.

The graph channel on its own

Used alone, the causal-factor graph reached nDCG@5 0.713, the highest of all systems, and was clearly better on Genesis, Challenger, WIRE and Ariane 501. P-graphonly − P = +0.076 (95% CI [+0.011, +0.146], p = 0.06; not significant after Holm correction). Interpretation: fused as one of seven lists, the graph signal is diluted. This is a hypothesis for the full study, not a conclusion.

Generated root-cause drafts (10 pilot queries)

System	Valid JSON	Items	Citations	Invalid citations	Uncited items
B0	10/10	57	0	0	57
B3	10/10	65	141	0	7
P	10/10	86	379	0	0

P cites evidence for every item, but **all 43** of its cause and factor statements are taxonomy category names ("Procedure deficient or not followed", "O1 Procedure deficient or not followed") instead of specific mechanisms; B3 had none. (An earlier hand count of 21 missed the code-prefixed items.) Citation validity is not correctness. The fix (P2) is in chapter 14.

Abstention (questions about missions not in the corpus, synthetic question set)

System	Abstained on 10 unanswerable questions
B0 (no documents)	0 / 10
B3	8 / 10
P	9 / 10

Audit

All pilot runs passed the independent audit: no own-mission leakage, no query copied from the reports (the owner's descriptions are original), every retrieved passage verified against its page, and every metric value recomputed independently and found equal.

14. Review-Driven Improvements (P2)

The guide was reviewed by two external AI assistants (Gemini, ChatGPT). Useful suggestions were adopted, but only after **freezing the design in writing before running it** (protocol section 9, commit e999e42), so that nothing was tuned on the pilot results. A suggestion that amounted to picking whatever scored best on the pilot was declined.

Change	Specification
P2: weighted graph channel	Graph list gets RRF weight 6 (= the six text lists together)
Generic-code filter	Drop expanded query codes present in at least 80% of missions (removes T4)
Specific-mechanism generation	Prompt demands the specific failure and how it led to the mishap; bare category names are dropped
Support check	LLM check that each quote supports its claimed factor, plus a 150-item human review sheet (D4)
Bias diagnostic	Label-free test of whether systems favour missions with long reports
Held-out reporting	Final results also on the 12 missions not used in the pilot

Retrieval (pilot, AI-assisted labels)

System	nDCG@5	nDCG@10	Recall@5	MRR	NormRecall@100
B3	0.648	0.800	0.580	0.733	0.019
P	0.638	0.816	0.540	0.833	0.023
P2	0.678	0.847	0.545	0.883	0.055
P2-nofilter	0.677	0.848	0.534	0.883	0.047
P-graphonly	0.713	0.860	0.598	0.883	0.056

Comparison (nDCG@5)	Difference	95% CI	p	p (Holm)
P2 - B3	+0.030	[-0.091, +0.148]	0.65	1.00
P2 - P	+0.040	[-0.035, +0.120]	0.37	1.00
P-graphonly - P	+0.076	[+0.011, +0.146]	0.06	1.00

Weighting the graph channel moves P towards the graph-only result; none of the differences is significant on 10 queries. The primary comparison remains P vs B3.

Bias towards long reports (no labels needed)

System	Correlation with report size	Top-5 rate: full reports	Top-5 rate: short documents
B1	+0.62	0.34	0.09
B2	+0.44	0.34	0.09
B3	+0.56	0.37	0.06
P	+0.45	0.36	0.07
P2	+0.21	0.27	0.18
P-graphonly	+0.03	0.21	0.25

The text-based systems mostly rank missions with long reports in the top 5, whatever the query; the graph channel does not (correlation near zero), and P2 reduces the bias. Part of the text

systems' scores may reflect report length rather than causal similarity; the human benchmark will test this with results split by document type. (This corrects an earlier statement that the analogy score favours long reports: pairwise scores do, rankings do not.)

Specific answers

System	Cause/factor items	Only a category name	Kept
B3	31	0	31
P	43	43 (100%)	43
P2	41	28 (68%)	**12, all specific**

The prompt alone removed only part of the generic answers; the enforcement removed the rest, leaving specific mechanisms such as "Frangible joint on one of the forward side fairing rails did not fracture". The cost is shorter answers. A two-step design (extract the mechanism from the cited passage first) is the next candidate.

Does the quote support the claimed cause?

The local LLM judged that **1,712 of 2,038** verified factors (84%) are supported by their own quote. This is a weak self-check (same model family as the extractor); the human verdicts in sheet D4 (100 accepted + 50 rejected factors) are the real measurement.

15. Key Findings and Discussion

1. **The base method transfers to spaceflight reports** and is reproducible, but its fusion is sensitive to field size: a small recommendation field dominated the results until weights were balanced.
2. **Verification matters.** 27% of the LLM's proposed causal factors were rejected; 504 cited "quotes" did not exist in the passage — exactly the hallucination risk the method guards against.
3. **No significant gain yet.** On the pilot, P matched B3. The causal-factor graph alone ranked best, which points to a better way of combining the graph with text retrieval.
4. **Grounded answers can still be generic.** P never cited a missing passage, but answered only with category names; P2's prompt plus enforcement leaves only specific mechanisms.
5. **Abstention works:** only the retrieval-grounded systems declined questions outside the corpus.
6. **Local LLMs are not reproducible across sessions;** caching outputs is necessary for honest reporting.

Proposed improvements before the full study (to be decided before human labelling)

- Add the graph-only or a weighted graph list as a pre-registered secondary system.
- A generation prompt that requires specific mechanisms rather than category names.
- Down-weight generic categories (O1, T1, T5) that appear for most queries.

16. Limitations and Honest Scope

- **Labels:** the pilot labels are AI-assisted; the synthetic proof-of-concept labels are keyword rules. Neither is a human benchmark.
- **Sample size:** 10 pilot queries (22 in the full design) — only large effects can be detected.
- **Field maps** are drafts pending human review.
- **Corpus:** 10 of 23 primary documents are summaries, press releases or slides, much shorter than full reports. The text-based systems strongly favour missions with long reports (chapter 14); the graph channel does not.
- **Models:** an 8-billion-parameter local model generates, extracts and judges; the automatic answer judge proved too lenient (it gave 0.81 key-point recall to the system that sees no documents).
- **Claims:** the project does not claim that P outperforms the base paper.

17. Project Status: Done and Remaining

Done

Part	Status
Corpus, ingestion, provenance (27 documents, 22 missions)	Done, verified
Base-paper reproduction (B0–B3, B3-bal)	Done, reproducible
Proposed method P and 6 ablations	Done, unit-tested
Evaluation, statistics, independent audit	Done; audits PASS
Annotation kit and guide	Done
Synthetic proof of concept	Done (not reportable)
10-mission pilot with AI-assisted labels	Done; results in chapter 12
Stage reports, submission summary, this guide	Done
Review-driven improvements (P2, bias diagnostic, support check, D4 sheet)	Done (chapter 14)

Remaining (after the project submission)

Task	Who	Effort
Decide the three design improvements (chapter 14)	Owner	1 h
Review taxonomy and field maps	Owner	2–3 h
Full human labelling: D2a, D2q, D2b, D1, D3 + second annotator	Owner (+ 1 person)	60–80 h + 12 h
Rate 40 blind answers (validate the automatic judge)	Owner	2 h
Review 150 extracted factors (sheet D4)	Owner	3 h
Final runs, audit, fill the IEEE paper's results	Assistant	about 2 h compute
Final secret scan, push to GitHub, pull request into main	Together	at the end

18. How to Run the Project

Setup

```
pip install -r requirements.txt
python -m spacy download en_core_web_sm
ollama pull llama3
copy .env.example .env      (set NEO4J_URI, NEO4J_USERNAME, NEO4J_PASSWORD)
```

Also required: Tesseract and Poppler (OCR), a Neo4j 5 database, and the bge-m3 model (downloaded once).

Pipeline

Step	Command
Parse documents	`python src/pipeline/page_parser.py --run-id ID`
Validate provenance	`python scripts/validate_passages.py --run-id ID`
Apply field maps	`python scripts/field_map_tool.py apply`
Load passages into Neo4j	`python scripts/ingest_passages_run.py ID`
Extract causal factors	`python scripts/extract_factors.py --run-id ID`
Build factor graph	`python scripts/build_factor_graph.py --run-id ID`
Run all systems	`python scripts/run_retrieval.py --run-id ID --queries FILE --proposed P P-graphonly ...`
Evaluate	`python scripts/evaluate.py --run runs/ID --eval-id EID [--labels DIR --pilot]`
Audit	`python scripts/audit.py --run runs/ID --eval runs/EID --out runs/AID`
Check annotation sheets	`python scripts/check_annotations.py [--dir DIR]`
Tests	`python -m pytest`

Every run writes its configuration, environment (code version, model versions), predictions, metrics, timings and log to `runs/<run_id>/`; existing run folders are never overwritten.

19. Guide to the Remaining Human Labelling

The full guide is `evalutation/benchmark/ANNOTATION_GUIDE.md`. In short:

1. **Review the taxonomy** (causal_taxonomy.json) and freeze it.
2. **D2a:** for each of the 22 missions, record 3–10 causal factors the board states: code, role, PDF page, a short verbatim quote. At least one root factor per mission.
3. **D2q:** for each mission, write 100–200 words describing the mishap as first observed — no causes, no copied text.
4. **D2b:** grade all 231 mission pairs 0–3 from the D2a factors.
5. **D1:** mark the causal factors stated in each of 200 sampled passages.
6. **D3:** write about 105 questions (single-mission, cross-mission, lessons, unanswerable) with key points and evidence pages.
7. A second person repeats a subset for agreement ($\kappa \geq 0.6$ targeted).

> Rules: record only what the board states; do not look at any system output while labelling; do not change the method after seeing labels. Run the checker after each step.

20. Glossary

Term	Meaning
BM25	Classic keyword ranking: rewards rare words that appear often in a passage
Dense retrieval / bge-m3	Ranking by meaning: text is turned into vectors and compared by similarity
RRF	Reciprocal Rank Fusion: merges ranked lists by summing 1 / (60 + rank)
Field	Section type of a report: Summary, Causes, Disposition (recommendations)
Causal factor	Something the board states caused or contributed to the mishap
Verification	Keeping a factor only if its quote is found word-for-word in the report
Factor graph	Neo4j graph linking missions to the causal-factor categories they share
nDCG@5	Ranking quality of the top 5 results, from 0 to 1; higher is better
Recall@5 / MRR	Share of relevant missions found in the top 5 / 1 ÷ rank of the first relevant one
Leave-one-mission-out	When a mission is the query, its own reports are hidden from the search
Ablation	A version of the method with one part removed, to see what that part contributes
Permutation test, CI	Checks whether a difference could be chance; range of plausible true differences
Abstention	The system answers 'insufficient evidence' instead of guessing
Pilot	A small first evaluation (here: 10 missions) to test the full procedure

21. References

1. S. Kim and S. Kim, "Multi-field hybrid retrieval-augmented generation for maritime accident root cause analysis," in *Proc. IEEE CASE*, 2026, pp. 238–243, doi:10.1109/CASE69030.2026.11704480.
2. X. Xiang, X. Chen, and J. Yang, "Knowledge graph-based causal analysis of aviation accidents: A hybrid approach integrating retrieval-augmented generation and prompt engineering," *Aerospace*, vol. 13, no. 1, Art. 16, 2026, doi:10.3390/aerospace13010016.
3. Y. Yang, G. Chen, B. He, and Y. Zhao, "Grounded knowledge graph extraction via LLMs: An anchor-constrained framework with provenance tracking," *Computers*, vol. 15, no. 3, Art. 178, 2026, doi:10.3390/computers15030178.
4. X. Xie, X. Tang, S. Gu, and L. Cui, "An intelligent guided troubleshooting method for aircraft based on HybirdRAG," *Sci. Rep.*, vol. 15, Art. 17752, 2025, doi:10.1038/s41598-025-02643-2.
5. G. V. Cormack, C. L. A. Clarke, and S. Buettcher, "Reciprocal rank fusion outperforms Condorcet and individual rank learning methods," in *Proc. SIGIR*, 2009, pp. 758–759.
6. S. A. Shappell and D. A. Wiegmann, "The human factors analysis and classification system — HFACS," DOT/FAA/AM-00/7, 2000.
7. Presidential Commission on the Space Shuttle Challenger Accident, *Report to the President*, Vol. 1, 1986.
8. Columbia Accident Investigation Board, *Report*, Vol. 1, 2003.
9. Mars Climate Orbiter Mishap Investigation Board, *Phase I Report*, 1999.
10. Ariane 501 Inquiry Board, *Ariane 5 Flight 501 Failure — Report by the Inquiry Board*, ESA, 1996.